



Activity File: Network Addressing

In this activity, you will continue to play the role of a security analyst at Acme Corp.

- Your security manager identifies the potential hacker that works at Acme Corp. Your manager is also capturing the potential hacker's traffic. They believe the hacker is accessing secret internal resources.
- While your security manager is able to capture the server IP address the hacker is accessing, they are only able to capture the raw binary traffic.
- Your task is to convert the binary traffic into an IP address, determine if it is a public or private address, and compare the IP to a list of Acme's servers to see which system the hacker is attempting to access.

ACME CORP SERVER LIST

IP RANGE	TYPE OF SERVER
10.8.42.32/24	Press Releases Server
10.6.0.0/28	Payroll Server
10.2.0.0/28	Payroll Server
10.3.0.0/28	Payroll Server
10.4.0.0/28	Payroll Server
10.5.0.0/28	Payroll Server
10.0.0.0/24	Trade Secrets Server
192.168.0.1/28	Payroll Server
192.168.0.2/28	Payroll Server
192.168.0.2/29	Human Resources Server
192.168.0.3/28	Human Resources Server
192.168.0.4/28	Human Resources Server
192.168.0.5/28	Human Resources Server
192.168.0.5/28	Human Resources Server
192.168.0.6/28	Human Resources Server
192.168.69.0/24	Trade Secrets Server
192.172.69.0/24	Trade Secrets Server
41.45.182.0/26	Intellectual Property Secrets
192.10.0.1/28	Finance Server
192.11.0.2/28	Finance Server
192.12.0.2/29	Accounting Server
192.13.0.3/28	Accounting Server
192.14.0.4/28	Accounting Server
192.15.0.5/28	Accounting Server
192.16.0.5/28	Accounting Server

Instructions

1. Convert the following binary representations into numeric IP addresses:
 - i. 11000000101010000100010110010001
 - ii. 0000101000000000000000000101010

- iii. 11000000101011000100010110010001
 - iv. 00101001001011011011011000100000
 - v. 0000101000000000000000001001100
2. Determine whether the numeric IP addresses are public or private.
 3. Compare the numeric IP addresses to the Acme server list and determine to which server the IPs belong.
 4. Summarize your findings to determine what resources the hacker is trying to access.

Bonus

Your security manager also finds the following binary data in the hacker's logs:

- 100010001111011111000111011001010001101000110110

Determine what this binary data represents.

Hint: This may translate to an IP address, but it is not actually an IP address.